

RESEARCH



Selective resonance ionization of calcium odd isotopes with odd–even selection rules

Stephen R. Wells¹ · Yoshihiro Iwata^{2,3} · Masabumi Miyabe^{2,3} · Shuichi Hasegawa^{1,2}

Received: 7 October 2025 / Accepted: 6 February 2026
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Abstract

Odd-even isotope selectivity in calcium was investigated using a laser resonance ionization $J = 0 - 1 - 0$ transition scheme, chosen so that angular momentum selection rules could be applied using linearly polarized light. Suppression of the abundant isotope ^{40}Ca and selection of the rare isotope ^{43}Ca was confirmed as a function of the linear polarization angle, through simple application of a half-waveplate ($\lambda/2$). In the electric dipole basis, by the selection rules, setting transition polarizations linearly orthogonal forbids the excitation of the even isotopes, while allowing the odd isotope excitation. Spectroscopy in the Rydberg level transition also showed selectivity in the resonance ionization process, with a final maximal (dependent on hyperfine transition) selectivity reported as a separation coefficient $\beta = 9e3$. It is expected that this relatively simple method in increasing selectivity of resonance ionization will have application to separation of the rare odd Ca isotopes as they are important in areas such as quantum information via ion trapping (^{43}Ca), and cosmology and biomedicine (^{41}Ca).

1 Introduction

Isotope separation is by nature a difficult process, with chemical methods generally rendered unsuitable due to the near-identical chemical properties across isotopes of the same element. In particular, for the ultra-trace analysis of low abundance isotopes, separation techniques are required to be highly selective in the targeted isotope against any higher abundance isotopes that can interfere with the signal. Several isotopes of calcium (atomic number = 20) are of significant research interest. Beginning with the heaviest relevant isotope and working backwards ^{48}Ca has a doubly magic nucleus and is the lightest known isotope to undergo double beta decay, while also being of high interest

for the possible detection of neutrino-less double beta decay (see e.g. Ref. [1]). ^{43}Ca being an odd nucleon numbered isotope has nuclear magnetic moment $I = 7/2$, giving rise to hyperfine structure, the ground state splitting in $^{43}\text{Ca}^+$ ($4s\ ^2S_{1/2}\ F = 3 : 4$) of which is attractive for quantum bit storage due to the large (3.2 GHz) splitting interval making state storage and discrimination relatively easy [2–5]. ^{41}Ca is an extremely rare radionuclide having long half life ($t_{1/2} = (9.94 \pm 0.15) \times 10^4$ years [6]) and solely electron capture decay with subsequent low energy x-ray release [7–9]. It has many possible applications, notably in cosmology where its particular half life is ideal as a radionuclide ‘chronometer’ for solar system time scale investigations [10–12]. In terms of nuclear engineering, ^{41}Ca is produced as a by-product due to neutron activation of ^{40}Ca in the high-neutron flux environments of nuclear reactors. For decommissioning, the ^{41}Ca radioisotope content must be quantified for appropriate waste streaming [13–15]. With calcium being a major constituent of bones, there is significant potential for ^{41}Ca to be used as a biomedical radio-tracer, if practical hospital based analysis techniques could be developed [16].

The analysis of these isotopes is strongly hampered by the very large relative abundance of the isotope ^{40}Ca . By percentage the isotope abundances are: ^{40}Ca 96.94%; ^{42}Ca 0.65%; ^{43}Ca 0.14%; ^{44}Ca 2.09%; ^{46}Ca 0.004%;

✉ Shuichi Hasegawa
hasegawa@tokai.t.u-tokyo.ac.jp

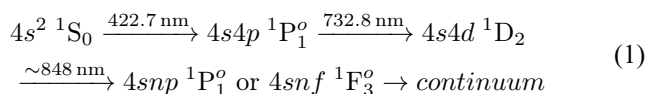
¹ Department of Nuclear Engineering and Management, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan

² Nuclear Professional School, The University of Tokyo, 2-22 Shirane, Shirakata, Tokai-mura, Ibaraki 319-1188, Japan

³ Collaborative Laboratories for Advanced Decommissioning Science (CLADS), Japan Atomic Energy Agency, 2-4 Shirakata, Tokai-mura, Ibaraki 319-1195, Japan

^{48}Ca 0.19% [17], with ^{41}Ca present in nature in only trace amounts. For the analysis of isotopes other than ^{40}Ca , a highly selective methodology is required.

Often employed for isotope selective investigations is resonance ionization mass spectrometry (RIMS). This ionization technique employs a number of lasers with wavelengths tuned to atomic resonances, so as to allow excitation of an electron in a stepwise ladder fashion, before being ejected from the atom. Since the particular resonances associated with the intermediate transitions have unique wavelength according to the nuclear mass and volume effect (known as the isotope shift), tuning the laser wavelengths to resonances of the isotope of interest allows only the excitation of that isotope. However, because both the laser and the atomic transition have finite linewidths rather than single values, some non-targeted isotope may also be ionized [18]. Investigations in RIMS for selective ionization of both ^{43}Ca and ^{41}Ca have been reported on [4, 19–21]. In the majority of these references, the RIMS procedure follows a first step resonance excitation at 422.7 nm, followed by either a single photon absorption: $4s^2\ ^1S_0 \xrightarrow{422.7\text{ nm}} 4s4p\ ^1P_1^o \xrightarrow{<390\text{ nm}} \text{continuum}$, or a double resonance excitation followed by non resonant excitation into the continuum: $4s^2\ ^1S_0 \xrightarrow{422.7\text{ nm}} 4s4p\ ^1P_1^o \xrightarrow{732.8\text{ nm}} 4s4d\ ^1D_2 \xrightarrow{<835\text{ nm}} \text{continuum}$, or a triple resonance excitation into Rydberg levels before ionization through electric field or additional photon absorption [22, 23]:



Since Scheme (1) is very commonly used in the literature, it will hereafter be referred to as the 'conventional scheme'.

While the conventional scheme has been proven to achieve very high optical selectivities (see e.g Ref. [24]) in this three-step resonance process, there are some drawbacks. Most significantly, non-resonant excitation with the additional absorption of a 422.7 nm or 732.8 nm photon from the $4s4d\ ^1D_2$ state reduces the selectivity as this excitation has enough energy to proceed without wavelength discrimination. In addition, in experiments such as these where the atom sample is produced by thermal vaporization, temperature dependent Doppler broadening will reduce selectivity.

Recently, interest in selectivity that can be attained between odd and even nucleon numbered isotopes via application of optical angular momentum selection rules has been renewed, after the technique was first discussed in the 1970 s [25–27]. A group at the RIKEN lab of Japan reports odd-even mass selectivity based on a resonance ionization scheme with laser polarizations set linearly orthogonal to

one another for isotopes of palladium [28–30]. By quantum optical selection rules, this forbids excitation in the even nucleon numbered isotopes while excitation in the odd isotopes remains allowed. They reported selectivity measurements of odd-mass Pd isotopes constituting > 99.7% of total isotope yield. Uses of this technique have also been reported in the past for isotopes Ba and Sr [31], Os and Ti [32], and more recently on Sm [33] and Zr [34].

Here, we investigate the odd-even polarization selectivity effect on isotopes of calcium using a newly investigated resonance ionization scheme.

2 Methods

The optical dipole moment matrix element between magnetic sublevels can be written as [35]

$$\begin{aligned} \langle J, F, M | d_{MM'} | J', F', M' \rangle \\ = \langle J | d_{MM'} | J' \rangle (-1)^{J+I+2F-M} \sqrt{(2F+1)(2F'+1)} \\ \times \begin{Bmatrix} J & F & I \\ F' & J' & 1 \end{Bmatrix} \begin{pmatrix} F & 1 & F' \\ -M & q & M' \end{pmatrix}, \end{aligned} \quad (2)$$

where J, F, M are the electronic angular momentum quantum number, atomic angular momentum quantum number and magnetic quantum number of the lower magnetic sublevel, respectively and J', F', M' are those of the upper magnetic sublevel. The reduced matrix element of the transition $\langle J | d_{MM'} | J' \rangle$ is only dependent on the electronic angular momentum quantum numbers. $\{\}$ and $()$ are the Wigner $6j$ and $3j$ symbol, respectively. I is the nuclear spin quantum number. The relationship between these quantum numbers can be written as $F = J + I$. q takes the values $-1, 0, +1$ depending on whether the transition light is left-handed circular, linear, or right-handed circular polarized light respectively. This part controls the optical transitions using the polarization of the laser beams. For the following discussion it may be helpful to refer to Fig. 1, as it shows a diagrammatic representation of the energy level structure and selection conditions. The quantization axis is defined to be parallel to the linear polarization electric field of the first laser beam. In order to take advantage of the polarization requirement, we set the first transition laser coupling the ground and first excited states ($4s^2\ ^1S_0 \rightarrow 4s4p\ ^1P_1^o$), to have linear polarization i.e. $q = 0$. Therefore the following selection rules apply; Equation (3) applies in the even isotope, zero nuclear magnetic moment $I = 0$ case.

$$\Delta J = 0, \pm 1 \text{ and } \Delta M_J = 0 \quad (3)$$

$$\Delta F = 0, \pm 1 \text{ and } \Delta M_F = 0. \quad (4)$$

Fig. 1 Energy level diagram showing the forbidden transition condition dependent on polarization orientation for the even isotope case shown in (a), and the allowed transitions in the odd isotope case, shown in (b). Note that the arrow lengths are not drawn correctly in (b), but the first step arrows represent $\Delta M_F = 0$ transitions and the second step arrows represent $\Delta M_F = \pm 1$ transitions. Structures not to scale

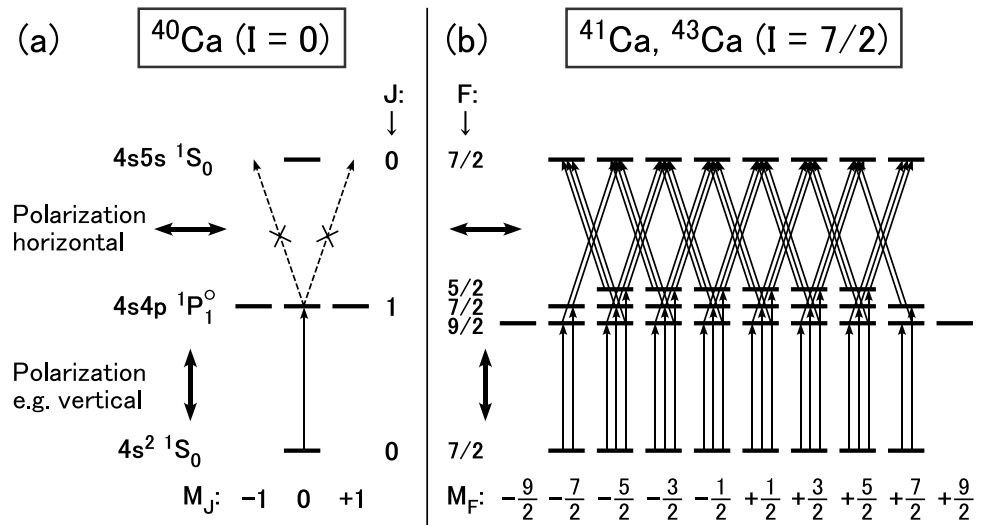
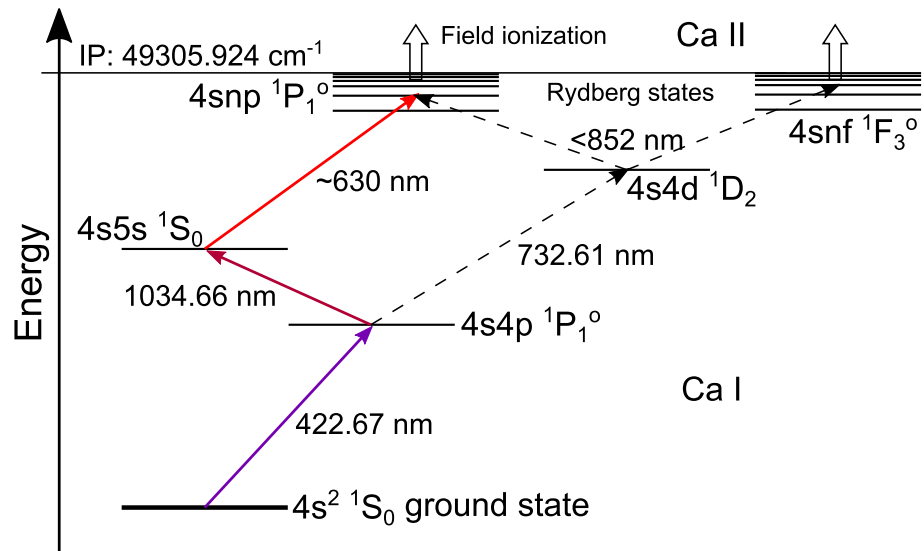


Fig. 2 Energy level scheme (not to scale) showing the resonance transition pathways taken for the new scheme (left, solid colored) and the conventional scheme (right, black dashes)



For calcium, the even isotopes have $I = 0$ and the odd isotopes ^{43}Ca and ^{41}Ca have $I = 7/2$. In the even isotope case, this first step excitation is able to proceed, as the $M_J = 0$ level is available and not forbidden. However, in the second step transition ($4s4p\ ^1P_1^o \rightarrow 4s5s\ ^1S_0$), setting the coupling laser polarization linear but orthogonal with respect to the first step laser, which sets the polarization basis for the transition scheme, changes q from $0 \rightarrow \pm 1$ as the linear polarization can be considered as a superposition of left- and right-circularly polarized light. Therefore, the selection rule condition as applies to this transition is $\Delta M_J = \pm 1$. Since the second excited level term is 1S_0 with $J = 0$ and a single projection $M_J = 0$, no levels are accessible and the second transition is forbidden by these selection rules. Should the polarizations be set in a parallel orientation relative to each other, then $\Delta M_J = 0$ and the transition would be allowed.

In the odd isotope case, with the restriction of $\Delta M_F = \pm 1$, some transitions become forbidden, for

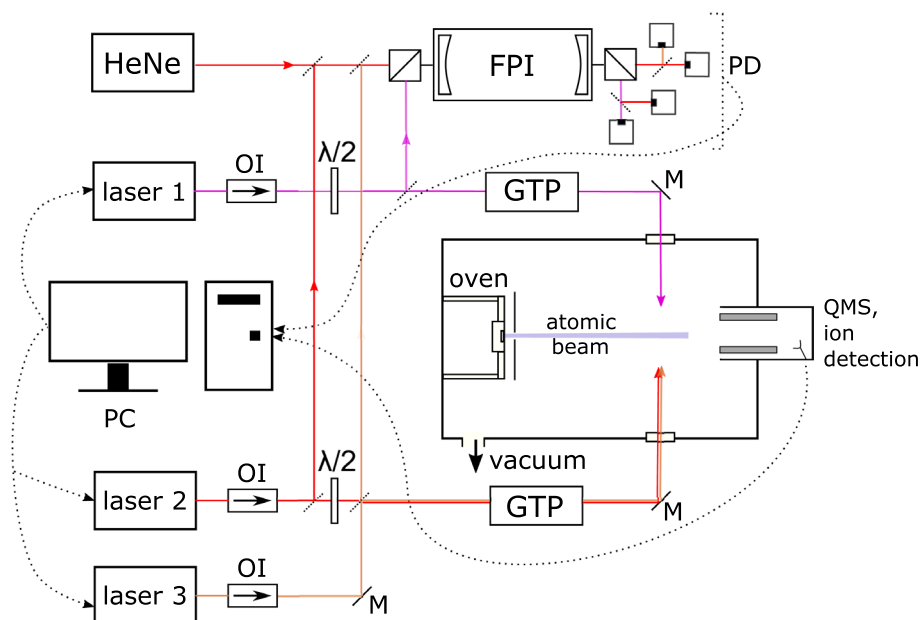
example with the level $^1P_1^o (F = 5/2)$ and assuming basis where $\Delta M_F = +1$, transitions from $M_F = -5/2$ to $M_F = -5/2, -7/2$ are forbidden. However, there remain a large number of transitions allowed, and so the transition can take place in the odd isotope case, with the caveat that the transition efficiency will be affected.

In order to access this $J = 0 \rightarrow 1 \rightarrow 0$ transition scheme, a newly investigated resonance ionization pathway was investigated. The new pathway, alongside the conventional scheme, is displayed in Fig. 2, and the experimental set-up is shown in Fig. 3.

The scheme is described in (5), with transition wavelengths (vac) $\lambda_1 = 422.67\text{ nm}$, $\lambda_2 = 1034.66\text{ nm}$ and $\lambda_3(n = 29) = 631.33\text{ nm}$:

$$4s^2\ ^1S_0 \xrightarrow{\lambda_1} 4s4p\ ^1P_1^o \xrightarrow{\lambda_2} 4s5s\ ^1S_0 \xrightarrow{\lambda_3} 4snP\ ^1P_1^o \xrightarrow{\text{field ionization}} \text{continuum} \quad (5)$$

Fig. 3 Schematic diagram showing the experimental setup for the ionization and polarization control investigation. Laser 1 outputs at 422.7 nm, Laser 2 outputs at 1034.66 nm, Laser 3 outputs at 631.33 nm, OI are optical isolators, FPI is a Fabry Perot Interferometer, GTP is a Glan Thompson prism, dotted lines in the laser beam paths indicate beam splitters, M indicates mirror, and QMS is quadrupole mass spectrometer



All the laser transitions are provided by homemade External Cavity Diode Lasers (ECDLs), with typical cw output powers: $\lambda_1 \sim 1$ mW; $\lambda_2 \sim 25$ mW; and $\lambda_3 \sim 30$ mW. With these cw lasers and beam spot size ~ 0.2 cm² (defined as the $1/e^2$ area) having relatively low intensity, non-isotope selective processes such as multiphoton ionization and tunneling ionization are effectively prevented, as might be the case in high-intensity pulsed laser regimes used to obtain higher ionization efficiencies. The beams cross orthogonally an atomic Ca vapor produced by electrothermal heating of a dried liquid sample inside a vacuum chamber. The first step laser beam is inserted in counterpropagation with the second and third beams to reduce Doppler broadening. Any ions that are produced are detected and identified with a quadrupole mass spectrometer (QMS). For polarization orientation control, the first and second step laser beams are passed through a half-waveplate, the rotation of which directly rotates the linear polarization orientation. Since ECDLs are known to not typically have 100% linear polarization purity, the beams are also passed through a Glan-Thompson prism with 100000 : 1 extinction ratio, to ensure elliptical polarization components are eliminated as much as possible.

Each experimental run is prefaced by optimization of the ion signal via iterative adjustment of the laser beam path, and frequencies. Since the ion yield decreases over time due to sample loss, the oven current is monitored to ensure yield consistency, with current increasing over the experimental day to obtain this.

The ECDLs are frequency stabilized and controlled by digital frequency offset lock (DFOC), via a reference cavity and stable He-Ne laser [14]. Using this control system, each laser has frequency stabilised and locked to the resonance

transition frequency of each step, with scanning capabilities available for spectroscopic investigation.

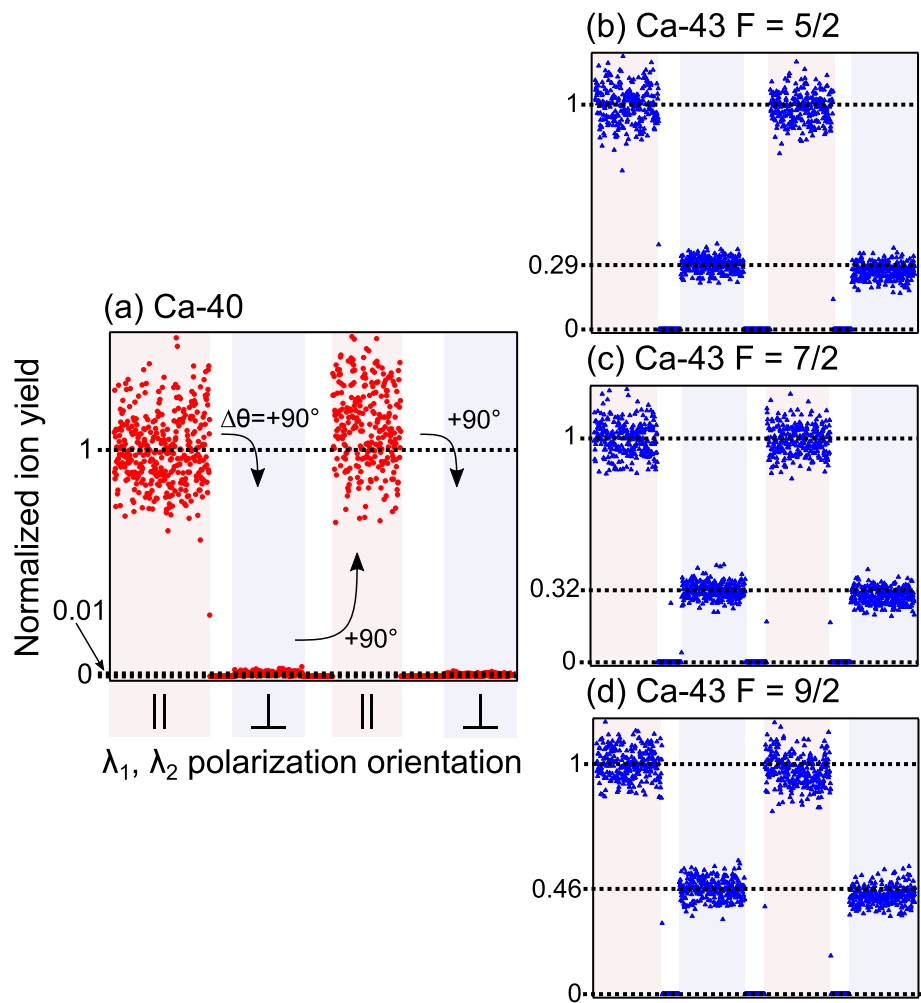
3 Results and discussion

As a first confirmation of the selectivity effect on the ionization yield of ⁴³Ca, the second step linear polarization was rotated through 360 ° and the yield recorded. A variety of methods of quantifying selectivity are reported in literature, and so in an effort to report the selectivity in a standardized manner, we use the 'selection factor', in simple form $\beta = \frac{A_1}{B_1} \cdot \frac{B_2}{A_2}$ [36]: the ratio of the target element (isotope) to the interfering element (isotope) in the sample after some selective process (A_1/B_1), divided by the same ratio in the sample before the selective process (A_2/B_2). In the case of calcium, since ⁴⁰Ca has extremely high abundance, and the other stable (even) isotopes have relatively small abundance, we approximate B_1, B_2 to be the measured ion yields of ⁴⁰Ca, rather than those of a summation of the naturally occurring isotopes. Here, A_1, A_2 is those of ⁴³Ca.

For clear investigation of the selectivity effect, a calcium sample with ⁴³Ca enriched was used. Sample abundances are: ⁴⁰Ca 23.8%, ⁴²Ca 1.0%, ⁴³Ca 62.2 ± 0.3%, ⁴⁴Ca 12.8%, ⁴⁶Ca < 0.01% and ⁴⁸Ca 0.2%.

Figure 4 shows the representative results of this polarization rotation experiment: ⁴⁰Ca⁺ in (a), and ⁴³Ca⁺ in (b–d). The maximum ion yield for the polarization condition set parallel is used to normalize the yield values, since ion yields are highly variable due to sample amount and oven temperatures. Due to the hyperfine structure of the $4s4p\ ^1P_1^o$ term of ⁴³Ca, three possible hyperfine levels

Fig. 4 Calcium normalized ion yields with respect to the first (λ_1) and second (λ_2) step transition laser linear polarization orientations. Part **a** shows the ion yield profile of $^{40}\text{Ca}^+$. The parallel condition regions are shaded lightly in a red color, while perpendicular condition regions are colored blue. To change from parallel to perpendicular and back, the second step $\lambda/2$ waveplate is rotated through an angle of 90° as indicated by $\Delta\theta$. To ensure ‘clean’ ion yields for each setting of the polarization condition, the lasers were completely cut while the $\lambda/2$ waveplate was rotated: this causes the regions of zero yield signal. Parts **b–d** show the ion yields for $^{43}\text{Ca}^+$ with respect to the same polarization conditions as in (a), with the same axes and polarization colorations. The hyperfine transition pathway is $F = 7/2 \rightarrow F'' \rightarrow 7/2 \rightarrow 9/2$ where F'' is $\frac{5}{2}, \frac{7}{2}, \frac{9}{2}$ for (b), (c) and (d) respectively



are available for the transition pathway: $F = \frac{5}{2}, \frac{7}{2}, \frac{9}{2}$ and so each is investigated as in Fig. 4b–d to assess the effect on ion yields due to choice of F level. However, the $4s29p^1P_1^\circ(F = 9/2)$ hyperfine level was used in each case as spectroscopic scans (as in Fig. 5) confirmed this principle transition ($J' = J + 1, F' = F + 1$) had maximal yield.

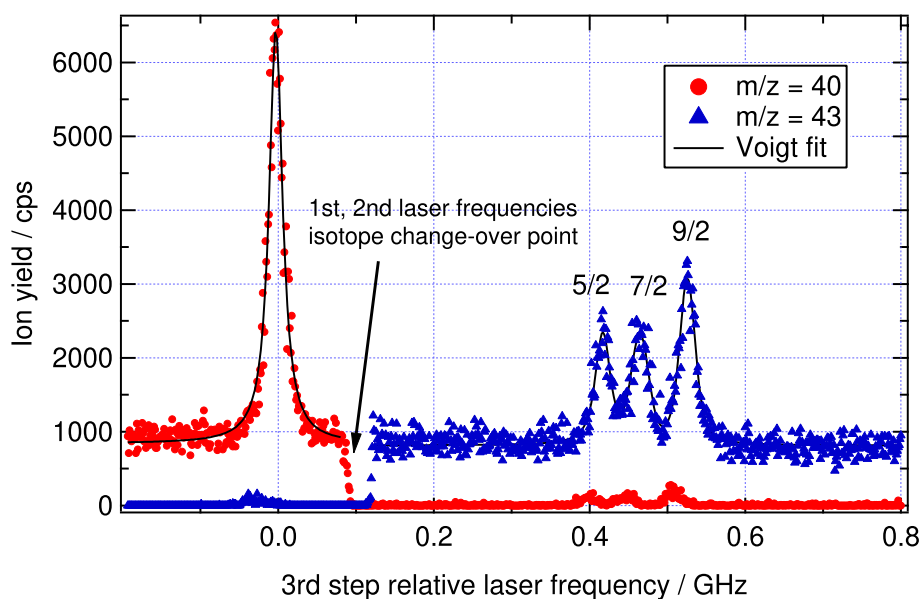
Figure 4a shows the effective suppression of $^{40}\text{Ca}^+$ with a rotation in the second step laser polarization orientation with respect to the first step. The limitations to achieve higher suppression of $^{40}\text{Ca}^+$ would be due to birefringence of the vacuum chamber windows. Another possibility is stray magnetic fields in the laser irradiation regions. Reference [37] have performed the polarization selected excitation and ionization experiment with Gadolinium. Since the Hanle effect affects the isotope selectivity of this methodology, they investigated the dependence on the magnetic fields. They have cancelled the magnetic field generated by the electron beam magnet and terrestrial magnetism by the compensation magnet, which gave them lower than 0.01 mT and good extinction of even isotope ion of Gd. This

magnetic field effect should be investigated to obtain better selectivity of our scheme.

Note that this $^{40}\text{Ca}^+$ value was taken for the laser frequencies optimized for ^{40}Ca ionization. Setting the laser frequencies to optimize the $^{43}\text{Ca}^+$ signal allows for additional selectivity to be achieved due to the isotope shift, in a three step scheme this can be extremely high. As expected, some differences in the yields of $^{43}\text{Ca}^+$ with respect to hyperfine level are seen, with the $F = \frac{9}{2}$ pathway providing the largest remaining yield of ^{43}Ca . For practical isotope separation procedures, choice of this level would be optimal.

It is important to note that the polarization selectivity methodology applies based on whether the isotopes in question have zero (even nucleon numbered isotopes) or non-zero (odd nucleon numbered isotopes), and as such are effective for selecting ^{43}Ca against ^{40}Ca . However, if the isotope of interest is, for example, the trace isotope ^{41}Ca then this polarization technique cannot be used for discrimination against ^{43}Ca . That being said, since ^{43}Ca has relatively low abundance (0.14 %), there will only be a very

Fig. 5 Spectroscopic scan of the $4s29p\ ^1P_1^\circ$ Rydberg state (endpoint of the third laser step $\lambda = 631.33$ nm) showing the difference in resonance frequencies between ^{40}Ca ($m/z = 40$) and ^{43}Ca ($m/z = 43$). Solid lines are Voigt fit shapes to the ion yield data. ^{43}Ca transition pathway is $F = \frac{7}{2} \rightarrow \frac{9}{2} \rightarrow \frac{7}{2} \rightarrow \frac{9}{2}, \frac{7}{2}, \frac{5}{2}$. Area marked by arrow and label is where the laser wavelengths λ_1 and λ_2 are shifted from the resonance values for ^{40}Ca to those of ^{43}Ca in order to record both resonance peaks as a function of $\nu_3 = \frac{c}{\lambda_3}$.



small probability of cross-ionization if narrow bandwidth lasers are used for the resonance ionization, and additional separation procedures are unlikely to be required.

A spectroscopic scan, shown in Fig. 5, was undertaken in order to evaluate the selectivity obtainable using this principle. The third step laser was scanned across an interval of 1.0 GHz containing the resonance transition frequencies for ^{40}Ca and ^{43}Ca . Here, during the scan of the third step frequency, the frequencies of the first and second step lasers were manually adjusted to correspond with and observe the relevant isotope with resonance frequency in the area of the scan frequency. The selectivity obtainable from the isotope shift is then calculated from the yield ratio of $^{43}\text{Ca}^+$ at its resonant frequency to $^{40}\text{Ca}^+$ at the resonant frequency of $^{43}\text{Ca}^+$.

The hyperfine structure is well resolved, giving a center-of-gravity isotope shift value relative to ^{40}Ca for ^{43}Ca for this Rydberg level ($4s29p\ ^1P_1^\circ$) of 1382(7) MHz, calculated by addition of the transition shift (475(7) MHz) obtained via fitting of Fig. 5 with the level shift (907.073(58) MHz) of the lower $4s5s\ ^1S_0$ level, obtained from [38].

A significant issue however, was the appearance of some small peaks observed in the $m/z = 40$ signal underneath the $^{43}\text{Ca}^+$ hyperfine structure peaks and vice versa. As explanation, we suspect in the main that the quadrupole mass spectrometer used here has a relatively large ion misidentification rate, and so when a large ion yield is being measured, for example at resonant transition frequencies, a certain portion will be counted in the wrong mass channel. This explanation is justifiable given that the appearance of $^{43}\text{Ca}^+$ ions at the three step resonance frequencies for $^{40}\text{Ca}^+$ is effectively impossible. Consequently we feel justified in calculating the selectivity via the yield ratios

Table 1 Yield ratio (against $^{40}\text{Ca}^+$) and separation coefficient data for each of the transition pathways in the $4s4p\ ^1P_1^\circ$ state and using the $4s29p\ ^1P_1^\circ\ F = 9/2$ Rydberg level in ^{43}Ca

$4s4p\ ^1P_1^\circ$ F level	Polarization yield ratio	Isotope shift yield ratio	Ratio product	Separation coefficient β
5/2	22	428	9333	3571
7/2	25	228	5698	2180
9/2	40	578	23023	8809

as described above, with yield counts assessed with the anomalous peaks discounted—the yield of $^{40}\text{Ca}^+$ was measured as the average count in the 3rd step frequency region between $\sim -0.2 \rightarrow -0.4$ GHz.

The overall data for selectivity and final separation coefficient calculation is listed in Table 1. Note that the isotope shift yield ratio and therefore the ratio product are calculated from the raw ion yields of ^{43}Ca and ^{40}Ca . To evaluate the separation coefficient β , the ratio product must be divided by the $^{43}\text{Ca}/^{40}\text{Ca}$ composition ratio in the sample; that is $\beta = (\text{Ratio Product}) \times 23.8/62.2$.

As expected from the polarization yield and peak heights, the transition pathway in $F = 9/2$ shows the highest overall selectivity.

A number of limitations in the procedures undertaken here are known. Since an all diode-laser system was used to provide the resonance ionization lasers, high enough intensities to saturate all transitions were not possible. In particular, the transition into the Rydberg state has relatively low transition cross section, limiting the observable ion yield. Since only odd isotopes are excited and ionized in this scheme, more ionization of odd isotopes could theoretically give better selectivity. This possibility should be more investigated with a new laser system.

However, this system has a number of important merits. As noted in the Introduction, with the ‘conventional’ resonance ionization scheme in calcium, since the non-resonant excitation is not a selective process (see e.g. Ref. [23]), a limitation is placed on attainable isotope selectivity. Specifically, non-resonant excitation may proceed via additional absorption of a first (422.7 nm) or second (732.8 nm) step laser photon from the second excited level ($4s4d\ ^1D_2$ in the conventional Scheme (1), and $4s5s\ ^1S_0$ in the new Scheme (5)). However, in the new scheme, as well as prohibiting the even isotope excitation, non-resonant excitation is forbidden with setting of the orthogonal linear polarization condition. It should be noted that in order to obtain a measurement of the isotope shift in Fig. 5, the polarization configuration is set parallel, and a non-resonant excitation background signal is seen (~ 1000 cps). Since such a process is unavailable in the conventional scheme, this presents a significant selectivity advantage.

In addition, this method is extremely accessible, requiring no complex experimental setups or equipment. A significant increase in selectivity was demonstrated simply by the manipulation of the linear polarization orientation; the waveplates to perform such a technique being a standard piece of optical equipment.

4 Conclusion

A newly investigated resonance excitation transition scheme in calcium was employed in order to manipulate optical angular momentum selection rules and the isotope selective effects were measured. The selection rule regime was probed by setting the excitation lasers in the linear regime, and rotating the second transition step laser polarization orientation. Suppression of the even isotope ion production, and non-suppression of the odd isotope ion production was confirmed. Including the selectivity obtainable from the isotope shift, a maximal separation coefficient value of $\beta = 8809$ was obtained through choice of the $4s4p\ ^1P_1^o\ F = 9/2$ and $4s29p\ ^1P_1^o\ F = 9/2$ hyperfine transition pathway. In the future, increasing laser intensities to fully saturate the Rydberg state transition should allow for higher selectivities to be measured.

Acknowledgements This research was partly conducted with the support of the Nuclear Safety Research & Development Center, Chubu Electric Power Company Inc.

Author contributions S.R.W.: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing original draft. Y.I.: Conceptualization, Methodology, Validation, Writing—review and editing. M.M.: Conceptualization, Methodology, Validation, Writing—review and editing. S.H.: Conceptualization, Resources, Validation, Writing—review and editing, Supervision,

Funding acquisition, Project administration.

Funding Open Access funding provided by The University of Tokyo

Data availability Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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